

How LoRA Fine-Tuning Changes Factual Predictions Across Transformer Layers

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Abstract

Parameter-efficient fine-tuning methods such as LoRA can substantially improve factual prediction accuracy, yet how these changes propagate across transformer layers remains poorly understood. We study *where* in the forward pass LoRA writes factual knowledge by applying logit-lens analysis and causal activation patching to Pythia-410m fine-tuned on CounterFact. We distinguish four conditions ordered by how much prior signal the base model has: facts it already knows (*known*), facts whose correct answer appears in the top-5 but not top-1 with meaningful probability (*latent*), real facts with no top-5 signal (*unknown*), and entirely invented facts (*synthetic*). Causal activation patching reveals a strict ordering of encoding depth across the three conditions where LoRA was needed: latent facts flip the base model at layer 10 (median), unknown at layer 16, and synthetic at layer 19 out of 24. Logit-lens trajectories provide convergent but distinct evidence: LoRA shifts predictions *earlier* for conditions with latent signal (known -2.5 layers, latent -0.4) and *later* for novel facts (synthetic $+3.8$ layers). All gains come at a steep locality cost: Kullback-Leibler (KL) divergence on neighboring-fact predictions over the top-50 tokens ranges from 4.9 to 5.4 nats across all conditions, proving LoRA induces massive representational drift regardless of the knowledge type.

1 Introduction

Low-Rank Adaptation (LoRA; [Hu et al. 2022](#)) has become one of the dominant methods for adapting large language models to new tasks. By injecting small trainable matrices at each attention layer, LoRA achieves competitive accuracy while updating fewer than 1% of model parameters. However, the improvement in final-layer predictions tells us little about *how* the model arrives at those predictions.

A natural question is whether LoRA simply amplifies an answer that was already latent in the network, or whether it installs the answer from scratch in the final layers. Prior work on language adapters found that adapted predictions often shift toward the target predominantly in the final layers (?). Whether LoRA for factual knowledge exhibits the same late-layer pattern — and whether this depends on the novelty of the fact being injected — is unknown.

We ask: **How does LoRA fine-tuning on factual knowledge affect the evolution of predictions across transformer layers, and does this differ along a spectrum from facts the model already knows, to facts where the correct answer is latent but suppressed, to facts it has no signal for, to entirely invented facts?**

To answer this, we fine-tune Pythia-410m-deduped on a stratified subset of the CounterFact benchmark ([Meng et al., 2022](#)) and analyse the resulting model using logit lens ([nostalgebraist, 2020](#)), tuned lens ([Belrose et al., 2023](#)), and causal activation patching. We run patching at every training checkpoint (not only the final model) to track *when* causal structure emerges during training. We supplement this with a rank ablation ($r \in \{4, 8, 16, 32\}$) and a locality audit on 15,000 neighborhood prompts using Kullback-Leibler (KL) divergence.

Our main finding is that LoRA’s causal encoding depth forms a strict spectrum for the three conditions where LoRA was required: latent (median first-flip layer 10) $<$ unknown (16) $<$ synthetic (19) out of 24. This ordering is stable across all 11 training checkpoints and all four ranks tested; higher rank increases how many facts are encoded, but not where. A particularly clean diagnostic emerges from the direction of first-layer shifts: conditions with latent signal shift *earlier* after LoRA; conditions without shift *later*. Finally, locality evaluation proves that LoRA is not a

surgical intervention, inducing massive representational drift across all knowledge types.

2 Related Work

Parameter-efficient fine-tuning. Hu et al. (2022) introduce LoRA, which freezes the pre-trained weight matrix W and learns a low-rank decomposition $\Delta W = BA$ with $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$. The effective change to each layer’s output is $\alpha/r \cdot BAx$, where α is a fixed scaling factor. LoRA has been applied to language modelling, instruction following, and knowledge editing, and is the basis of most modern fine-tuning pipelines.

Interpreting intermediate representations. The logit lens (nostalgebraist, 2020) decodes intermediate hidden states through the unembedding head to obtain a vocabulary distribution at each layer, enabling inspection of how predictions evolve through the forward pass. Belrose et al. (2023) extend this with a tuned lens: a learned affine probe per layer that predicts the final-layer output distribution more accurately than the raw logit lens. We use both, reporting the logit lens as the primary metric to avoid conflating lens-specific effects with LoRA-induced changes.

Adapters and layer dynamics. ? study how adapter modules and LoRA affect the hidden-state trajectory, finding that adapted predictions shift toward the target predominantly in the final layers for language-transfer tasks. We test whether the same late-layer dominance appears for factual knowledge, and whether it holds across knowledge types.

Factual knowledge in LLMs. Meng et al. (2022) show through causal tracing that factual associations in GPT models are primarily stored in mid-layer MLP modules. Our activation patching adapts their methodology: we patch hidden states from the fine-tuned model into the base model to find the earliest layer at which the LoRA update causally determines the output.

3 Methodology

3.1 Model and Dataset

We use **Pythia-410m-deduped** (Biderman et al., 2023), a 24-layer decoder-only transformer with hidden size 1,024. Facts are drawn from the **CounterFact** benchmark (Meng et al., 2022),

which provides 19,728 subject–relation–object triples along with paraphrase prompts for each fact. We retain only facts whose target answer is a single token under the model’s tokenizer, yielding 17,606 usable facts.

3.2 Experimental Conditions

We define four conditions ordered by how much prior signal the base model has for the correct answer:

- **Known (rank 1):** The base model predicts the correct answer as its top-1 token; 1,236 facts qualify (7.0%).
- **Latent (rank 2–5, log-prob > -5):** The correct answer appears in the model’s top-5 predictions but is not top-1, *and* carries meaningful probability mass (answer log-prob > -5 , i.e. $p \gtrsim 0.007$; 2,802 facts; 15.9%). The representation is present but suppressed. The secondary log-probability gate excludes 35 rank-2–5 facts whose near-zero probability signals accidental ranking rather than genuine latent knowledge; those facts are reassigned to the unknown pool.
- **Unknown (rank > 5 , or rank 2–5 with log-prob ≤ -5):** The correct answer has no meaningful top-5 signal at baseline (13,568 facts; 77.1%). The model lacks the association entirely.
- **Synthetic:** Pseudo-entity facts created by pairing real relation templates (e.g. “{*subject*} was born in”) with a randomly generated subject name and a real object from that relation. By construction the base model has never encountered the triple.

This four-way split tests a spectrum hypothesis: if LoRA’s encoding depth is determined by how absent the knowledge is, we predict the strict ordering known $<$ latent $<$ unknown $<$ synthetic in causal first-flip layer. We stratify-sample 500 facts per condition and 200 synthetic facts, capping each relation at 25%, and train for 5 epochs (535 optimizer steps). Paraphrase prompts are held out and never used in training.

3.3 LoRA Fine-Tuning

We attach LoRA adapters ($r = 16$, $\alpha = 32$, dropout 0.05) to the `query_key_value` projections of all attention layers (1.57M trainable pa-

Table 1: Logit-lens results before and after LoRA fine-tuning (Pythia-410m-deduped, $r = 16$, 535 steps). *First layer Δ* = shift in mean first-appearance layer (negative = earlier). Para. = held-out paraphrase prompts (never seen in training).

Condition	Prompts	Acc@1		Mean log-prob		First layer Δ
		Base	LoRA	Base	LoRA	
Known	train	0.988	0.988	-1.62	-0.09	-2.5
Known	para.	0.312	0.809	-3.92	-0.71	-1.2
Latent (top-5)	train	0.002	0.968	-3.25	-0.20	-0.4
Latent (top-5)	para.	0.132	0.745	-5.03	-1.00	-0.7
Unknown	train	0.000	0.780	-7.38	-0.93	+1.2
Unknown	para.	0.054	0.421	-6.96	-2.88	+0.5
Synthetic	train	0.000	0.590	-8.84	-1.79	+3.8

rameters; 0.39% of total). Training uses the cross-entropy loss on the answer token only (prompt tokens masked), AdamW with learning rate 10^{-4} , batch size 16, and FP16 mixed precision, for 5 epochs (535 optimizer steps). We save adapter checkpoints every 50 steps.

3.4 Layer-wise Analysis

For each adapter checkpoint we run the full forward pass with `output_hidden_states=True` and apply the logit lens (frozen final layer-norm + unembedding head) to every intermediate hidden state at the last token position. We record acc@1, mean log-probability, and the first layer at which the correct answer appears as top-1 (*first-layer-of-appearance*). The *base model's* tuned lens is loaded from the AlignmentResearch hub and kept frozen across all variants, ensuring that any change in its readout is attributable to LoRA, not to changes in the probe.

3.5 Causal Activation Patching

For each fact and each layer $\ell \in \{1, \dots, 24\}$, we patch the LoRA model's residual stream after block ℓ into the base model via a forward hook, and record whether the base model's output flips to the target answer. The *first-flip layer* is the smallest ℓ at which the flip occurs. A first-flip layer near 0 means the answer was already causally present (LoRA only amplified it); a first-flip layer near 24 means the entire update lives in the final layers. We run patching at every training checkpoint to track how causal depth evolves during fine-tuning.

4 Results

4.1 Layer-wise Prediction Evolution

Table 1 shows the final-layer statistics for all four conditions. Figure 1 shows how LoRA's gain in

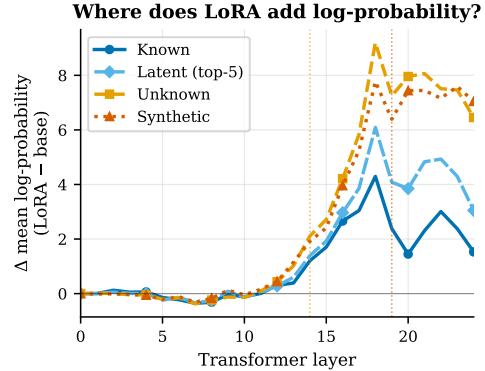


Figure 1: Mean increase in answer log-probability at each layer (LoRA final – base, training prompts). Vertical dotted lines mark median causal first-flip layers for unknown (layer 16) and synthetic (layer 19) from patching.

answer log-probability is distributed across the 24 transformer layers.

The four conditions show qualitatively distinct shapes that reflect their amount of prior signal.

Known facts show a moderate gain that rises gradually from early layers, consistent with shallow amplification. The first-layer-of-appearance moves 2.5 layers earlier ($21.2 \rightarrow 18.6$) and paraphrase accuracy jumps from 31.2% to 80.9%.

Latent facts (answer in top-5 but not top-1, and $\text{log-prob} > -5$) show the most striking result. Despite near-zero baseline accuracy (0.2%), LoRA raises training accuracy to 96.8% and held-out paraphrase accuracy to 74.5%. First-layer-of-appearance shifts only 0.4 layers earlier — the lightest structural change needed to unlock an already-present representation. This is the clearest evidence in our results that LoRA operates as a *representation unlocker* when the knowledge exists but is suppressed.

Unknown facts (no meaningful top-5 signal) show a large gain concentrated after layer 14, and the first-layer-of-appearance shifts 1.2 layers *later* — the first condition where the direction reverses. Paraphrase accuracy reaches 42.1%, substantially below latent (74.5%), reflecting the absence of prior structure to generalise from.

Synthetic facts show the most extreme late-layer spike pattern, with the first-layer shift reaching +3.8 layers later and training accuracy of 59.0%.

The direction of first-layer shift is itself a clean diagnostic: conditions with latent signal (known: -2.5, latent: -0.4) shift earlier; conditions with-

Table 2: Activation patching at the final LoRA checkpoint ($r = 16$, step 535). *Flipped* = facts (out of 100) where any single-layer patch causes the base model to output the target answer. The known condition is excluded: the base model already predicts correctly for those facts, so the result would reflect the selection criterion rather than LoRA’s causal footprint.

Condition	Flipped	Mean first-flip layer	Median
Latent	100/100	9.86	10
Unknown	82/100	15.30	16
Synthetic	64/100	18.25	19

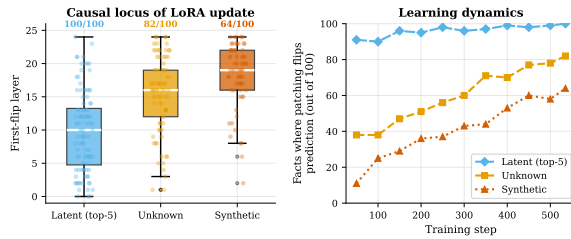


Figure 2: *Left*: Distribution of first-flip layers at the final checkpoint. Latent, unknown, and synthetic are cleanly separated; known is omitted from the causal analysis (see text). *Right*: Count of flipped facts across 11 training checkpoints (known omitted; trivially 100 throughout). Location is stable; count grows with training.

out it (unknown: +1.2, synthetic: +3.8) shift later. The boundary aligns exactly with the combined top-5 rank and log-probability criterion used to define the latent condition.

4.2 Causal Localization via Activation Patching

Table 2 and Figure 2 show the causal results for the three conditions where LoRA was required to produce the correct answer. By patching the full residual stream at each layer, we identify the earliest point at which LoRA-modified activations are sufficient to flip the base model’s output to the target. The strict ordering **latent** (median layer 10) < **unknown** (16) < **synthetic** (19) confirms the depth spectrum.

For **latent** facts, all 100 flip at a median of layer 10, placing the causal site in the mid-network region where Meng et al. (2022) identified factual associations in GPT. LoRA’s intervention here is minimal: it redirects an existing representation rather than writing a new one. For **unknown** facts, 82/100 flip at layer 16; for **synthetic** facts, 64/100 flip at layer 19.

The right panel of Figure 2 shows the train-

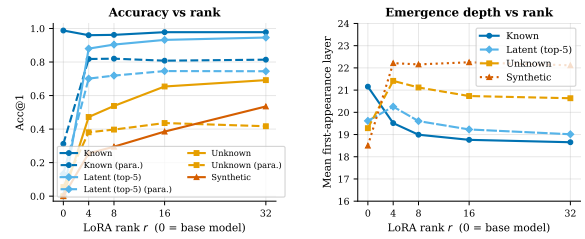


Figure 3: *Left*: Accuracy vs rank (solid = train, dashed = paraphrase). *Right*: Mean first-appearance layer vs rank.

ing dynamics across all 11 checkpoints (steps 50–535). The first-flip *layer* per condition is stable throughout: latent stays near 10, unknown near 16, synthetic near 19. What grows with training is the *count*. **Where LoRA encodes a fact is determined by the fact’s novelty from the very first training step; training only determines how many facts reach that threshold.**

4.3 Effect of LoRA Rank

Figure 3 confirms that rank governs capacity, not depth. For **synthetic** facts, train accuracy scales from 25.5% at $r = 4$ to 53.5% at $r = 32$, yet the mean first-appearance layer stays near 22 throughout ($r = 4$: 22.2, $r = 32$: 22.1). For **unknown** facts the pattern is similar (47.2% to 69.2% in accuracy), but depth drops slightly (21.4 to 20.6). For **latent** facts, accuracy is high at all ranks (88.0% to 94.6%) and depth decreases slightly with rank (20.3 to 19.0), consistent with larger adapters making slightly more efficient use of a pre-existing representation. **Known** facts are rank-saturated: train accuracy stays near 96–98% and paraphrase accuracy plateaus around $r = 8$ (82.0%), while their depth also drops (19.5 to 18.7).

The right panel makes the separation explicit: all real-world facts (known, latent, unknown) show a mild depth decrease of $\sim 0.8 - 1.2$ layers as rank increases from 4 to 32, taking advantage of the added capacity to surface answers earlier. Only synthetic facts stay strictly locked in the final layers (shifting just 0.1 layers). **Rank primarily acts as a capacity dial; the fundamental depth of extraction is still governed by the knowledge type.**

4.4 Locality of Factual Updates

Table 3 reveals an inverse relationship between distributional drift (KL) and top-1 preservation,

Table 3: Neighborhood locality across 5,000 related-fact prompts per condition. *KL div.* = mean $\text{KL}(P_{\text{base}} \| P_{\text{LoRA}})$ restricted to the top-50 tokens under the base distribution, with both distributions renormalized over that support. *Pres. rate* = fraction of prompts where LoRA top-1 equals base top-1.

Condition	KL div.	Pres. rate	N prompts
Known	5.45	0.223	5,000
Latent (top-5)	5.32	0.102	5,000
Unknown	4.92	0.063	5,000

highlighting a paradox of fine-tuning disruption. Known facts exhibit the highest top-50 KL divergence (5.45 nats) but also the highest top-1 preservation (22.3%). Conversely, unknown facts show the lowest KL divergence (4.92 nats) but the lowest preservation (6.3%).

This apparent paradox is explained by baseline confidence. Known facts have sharp, highly confident initial distributions; when LoRA perturbs the network, it scatters mass away from the dominant token, incurring an immense KL penalty. However, because the initial margin was so large, the top token often survives as the argmax. Unknown facts have flatter, more uniform initial distributions, so perturbations incur a smaller KL penalty but easily scramble the fragile top-1 ranking. Across all conditions, the absolute magnitude of drift (4.9–5.4 nats over just the top 50 tokens) confirms that LoRA induces massive collateral damage, regardless of depth.

5 Discussion

Our results establish a consistent spectrum across two independent measurements. Logit-lens analysis shows that LoRA’s first-layer-of-appearance shifts earlier for conditions with latent signal (known: -2.5 layers, latent: -0.4) and later for conditions without it (unknown: $+1.2$, synthetic: $+3.8$). Causal activation patching confirms a monotone ordering in absolute causal depth — but only for the three conditions where LoRA was actually needed: **latent** (median layer 10) < **unknown** (16) < **synthetic** (19) out of 24. The known condition is excluded from this causal ordering because its layer-0 result reflects the base model’s pre-existing capability, not LoRA’s encoding depth (see Section 4.2).

It is worth being explicit that these two measurements capture different things. The logit-lens shift is a *relative* change in the fine-tuned

model’s own internal trajectory: it measures how much earlier or later the correct answer *naturally emerges* in the LoRA model versus the base model, with no intervention. The causal first-flip layer is an *absolute* measure of causal sufficiency: the earliest point at which LoRA-modified activations, when injected into the *base* model, are sufficient to flip its output. The two metrics qualitatively agree — both put conditions with prior signal earlier and conditions without it later — but they are not equivalent, and their agreement should be read as corroborating convergent evidence, not as independent confirmation of a single underlying quantity.

The latent condition is the most theoretically informative. Latent facts yield an immense raw paraphrase generalisation gain ($\Delta = +0.613$). While this absolute delta is larger than that of known facts ($+0.497$), both conditions actually consume an identical proportion of their available accuracy headroom ($\sim 70\%$). Remarkably, latent facts achieve this elite generalisation with barely any structural change to the forward pass (-0.4 layers). This combination — large accuracy gain, maximal generalisation efficiency, and shallow causal site (layer 10) — points to LoRA acting as a *gating mechanism*: it adjusts attention weights so that a correct but suppressed representation wins the competition against the incorrect top-1 prediction. The log-probability gate used to define the latent category (answer log-prob > -5) reinforces this interpretation: the 35 reclassified facts that passed the rank-2–5 criterion but carried near-zero probability are semantically closer to unknown, and excluding them sharpens the latent cluster.

Unknown-fact fine-tuning, in contrast, must operate deeper (layer 16) and shows substantially weaker paraphrase generalisation (42.1% vs 74.5%), consistent with the model constructing a narrower, less compositional association. Synthetic facts, with no prior structure at all, can only be installed in the final layers (layer 19) and cannot be tested for paraphrase generalisation.

Together, these findings suggest that the late-layer dominance reported by ? for language adapters is real but incomplete: it applies to conditions where the model has no existing signal, while LoRA on latent or known facts operates substantially earlier. The important variable is not task type (language transfer vs. factual knowledge) but the degree to which the knowledge already exists

in the network.

A limitation is that all four conditions are trained jointly, meaning condition interactions cannot be disentangled without separate runs. Additionally, because synthetic subjects are generated pseudo-words, they are likely subjected to irregular subword fragmentation by the BPE tokenizer. This may induce out-of-distribution attention patterns compared to real entities, partially contributing to their delayed, late-layer extraction.

6 Conclusion

We have shown that LoRA fine-tuning for factual knowledge produces a consistent encoding-depth spectrum determined by prior knowledge, visible through both logit-lens trajectories and causal activation patching. The causal ordering spans the three conditions where LoRA was needed: latent facts (answer in top-5 with log-prob > -5) are encoded at median layer 10, unknown facts at layer 16, and synthetic facts at layer 19 out of 24. The known condition is a control: because the base model already answers correctly, its layer-0 patching result reflects the inclusion criterion, not LoRA’s causal footprint. The depth ordering is stable across all 11 training checkpoints and all four ranks tested ($r \in \{4, 8, 16, 32\}$): training increases how many facts are encoded, rank increases how many can be encoded, but neither changes where.

The latent condition is the practical takeaway: LoRA is most efficient when the correct answer is already in the model’s top-5 predictions *with meaningful probability* (log-prob > -5), delivering 96.8% train accuracy and 74.5% paraphrase generalisation — performing as well as known facts — from the smallest structural change. All conditions degrade locality severely: top-1 preservation rates range from 22.3% (known) down to 6.3% (unknown). We find that LoRA induces immense representational drift across all conditions, though sharp initial distributions (known facts) are paradoxically more robust to losing their argmax ranking despite incurring higher KL divergence penalties.

Practitioners who need high locality should prefer fine-tuning on facts where the model already has latent signal (top-5 and log-prob > -5 at baseline), monitor neighborhood KL-divergence during training as an early warning signal, and treat synthetic-entity injection as a last resort given its

combination of low generalisation and deep structural cost.

AI Disclosure and Reflection

We used Claude (Anthropic) and GitHub Copilot to assist with implementation debugging, data-pipeline design, and manuscript editing. The experimental design, hypotheses, analysis, and conclusions are our own. All reported numbers were produced by running our code on Google Colab A100 hardware and verified manually against the raw output logs.

References

- Nora Belrose, Zac Furman, Logan Smith, Danny Halawi, Tuu Oikarinen, Theodore Wu, Atticus Ziegler, and Jacob Steinhardt. 2023. Eliciting latent predictions from transformers with the tuned lens. *arXiv preprint arXiv:2303.08112*.
- Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, Kyle O’Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, Usven Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar van der Wal. 2023. Pythia: A suite for analyzing large language models across training and scaling. In *International Conference on Machine Learning (ICML)*.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. [LoRA: Low-rank adaptation of large language models](#). In *International Conference on Learning Representations (ICLR)*.
- Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. 2022. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- nostalgebraist. 2020. Interpreting GPT: the logit lens. <https://www.lesswrong.com/posts/AcKRB8wDpdaN6v6ru/interpreting-gpt-the-logit-lens>.